

PATIL ATHARVA PRAVIN

Machine Learning & Data Science Engineer

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PROFILE

Machine Learning Engineer and Data Science Specialist with hands-on experience in building end-to-end ML pipelines, predictive analytics, and unsupervised learning for real-world applications. Skilled in deploying ensemble models using XGBoost, Scikit-learn, Random Forest for classification, clustering, and demand forecasting on large-scale datasets. Proficient in exploratory data analysis, feature engineering, and geospatial visualisation using Pandas, NumPy, Matplotlib, Seaborn, and Folium. Experienced in cybersecurity-oriented ML applications including network intrusion detection using Scapy and packet analysis. Additionally skilled in full-stack web development using Node.js, Express, and MongoDB, enabling end-to-end system design from data pipeline to deployment.

WORK EXPERIENCE

Network Analyst Intern Dec 2025 – Jan 2026 Camlin Fine Sciences | Mumbai, India

- Assisted the security operations team in monitoring firewall activity and reviewing network traffic logs to identify potential anomalies and threat indicators.
- Gained practical exposure to SIEM workflows, alert triage, and network security monitoring tools in a production environment.
- Observed how cybersecurity monitoring and network security practices are applied in a live industrial setting.

PROJECTS

Multi-Model Intrusion Detection System

- Engineered a real-time network IDS trained on the NSL-KDD dataset (125,973 records, 41 features) using a weighted soft voting ensemble of Random Forest, SVM, and Logistic Regression, achieving 99.75% accuracy with 5-fold cross-validation of 99.68%.
- Implemented live packet capture and feature extraction using Scapy, mapping raw traffic into NSL-KDD format for real-time classification.
- Built a 3-level threat escalation pipeline (warning → alert → block) with automated IP blacklisting, timestamped attack logging, and a live Streamlit dashboard with auto-refresh.

Ride Demand Analysis & Forecasting

- Built an end-to-end data science pipeline on NYC ride-hailing data to forecast urban mobility trends and optimize fleet distribution.
- Performed EDA on temporal demand patterns across hours and days, fare analytics, and zone-level pickup rankings using Pandas, Matplotlib, and Seaborn.
- Applied KMeans unsupervised clustering on geospatial coordinates to identify 8 distinct demand hotspots, visualized via an interactive Folium heatmap with cluster centroids.

SeatSync — Intelligent Carpooling Platform

- Integrated a trust-scoring model using behavioral signals and ride history to rank driver reliability and flag anomalies, applying data-driven decision making to a real-world system.
- Designed fraud detection logic within the admin dashboard using rule-based and statistical heuristics on ride and payment data.
- Backend built with Node.js, Express, MongoDB, and Socket.io with JWT authentication, real-time geospatial route matching, and SOS emergency alerts.

EDUCATION

Haji Jamaluddin Thim Trust Theem College of Engineering

Bachelor of Engineering — Information Technology | Tarapur, India

Jul 2023 – Present

Tarapur Vidya Mandir

HSC — 57.33%

Jul 2022 – May 2023

C.B Sankhe School

SSC — 76.20%

Jun 2021 – May 2022

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS

ML / Data Science: Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Cybersecurity: Scapy, Wireshark, IDS/IPS, Network Sniffing, Packet Analysis

Frameworks: Node.js, Express, React.js, FastAPI, Streamlit

Tools: Git, GitHub, Postman, Google Colab, MongoDB, Socket.io

Domains: Machine Learning, Data Science, Predictive Analysis, Intrusion Detection, Anomaly Detection, Full-Stack Development

COMPETENCIES

- Analytical and problem-solving mindset
- Ability to learn and adapt quickly
- Good communication and presentation skills
- Ability to handle pressure in different circumstances
- Team collaboration and leadership
- Attention to detail in system design

DECLARATION

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.
Atharva Pravin Patil